8 **(a)** A teacher uses the arrangement shown in Figure 8.1 to demonstrate how an intruder alarm works.

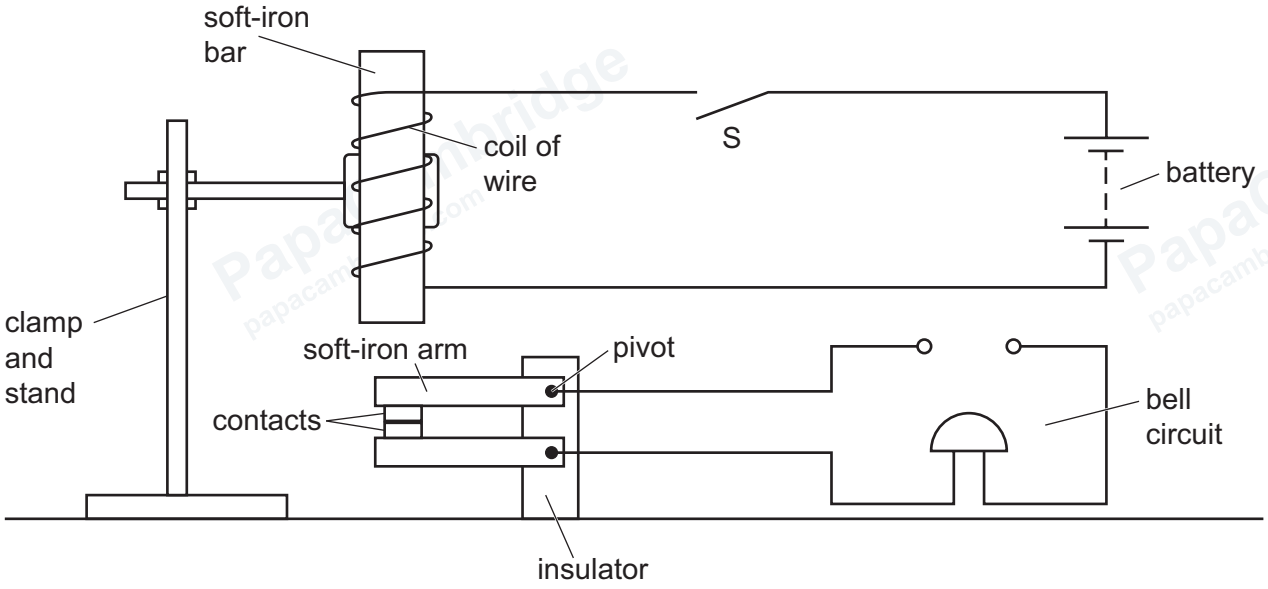

Figure 8.1

Switch S is connected to a door. When the door is open, switch S is open. When the door is closed, switch S is closed.

Complete the sentences to explain how the arrangement can be used as an intruder alarm.

The teacher closes the door and the battery produces a .....

in the coil of wire.

The soft-iron bar is ..... and attracts the soft-iron arm.

When the door is opened, the contacts touch and this .....

the bell circuit, causing the bell to ring.

[3]

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(b) Figure 8.2 shows a transformer that is 100% efficient.

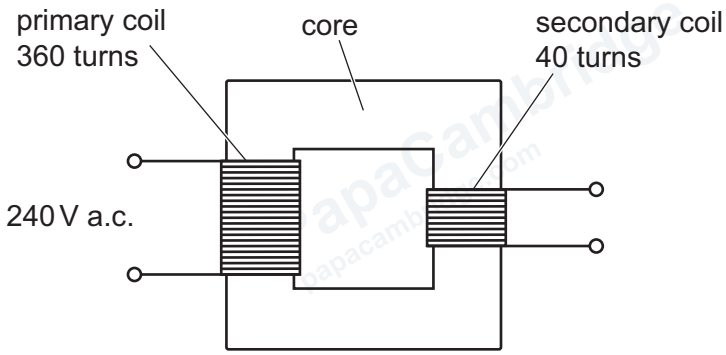

Figure 8.2

(i) Calculate the voltage V_s across the secondary coil.

V_s = ... V [2]

(ii) A student connects the primary coil of the transformer to a d.c. supply.

Explain why there is no voltage across the secondary coil.

.....

.....

..... [2]

[Total: 7]

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